

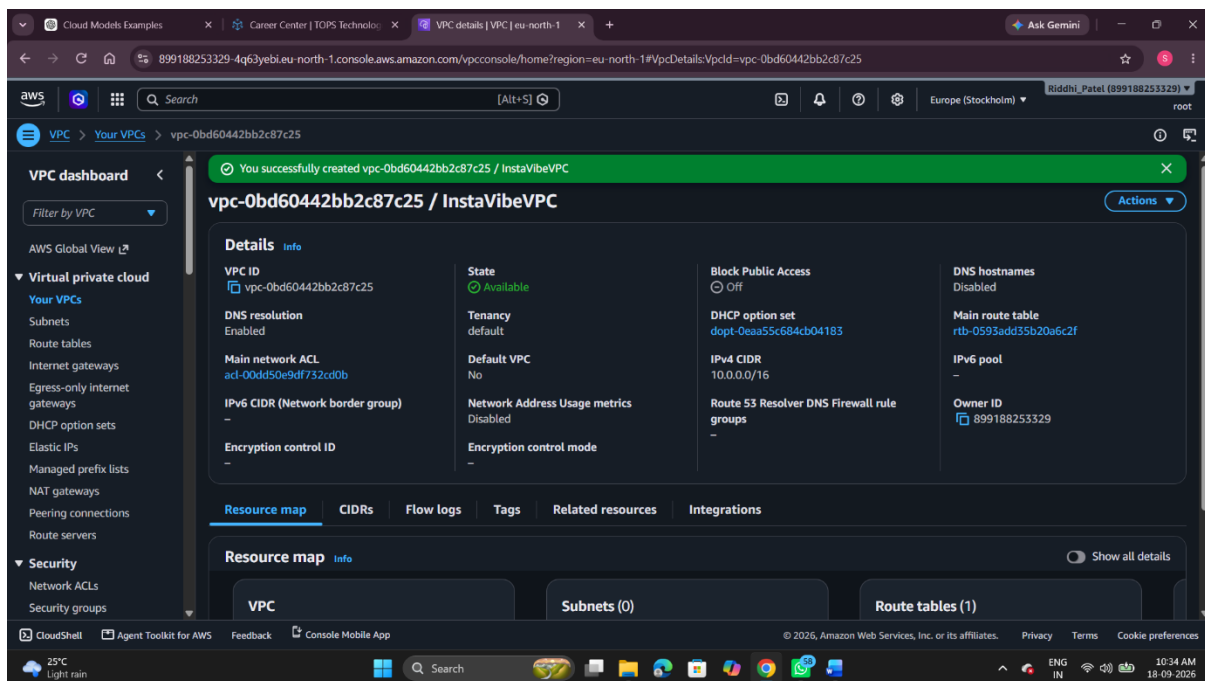
Cloud Networking – Assignment 1

1. Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).

Ans: Steps:

1. Log in to the AWS Management Console.
2. Open the VPC Dashboard.
3. Select Your VPCs → Create VPC.
4. Select VPC only.
5. Enter the name InstaVibeVPC.
6. Enter IPv4 CIDR block 10.0.0.0/16.
7. Keep tenancy as Default and click Create VPC.

Screenshot: InstaVibeVPC created with CIDR 10.0.0.0/16



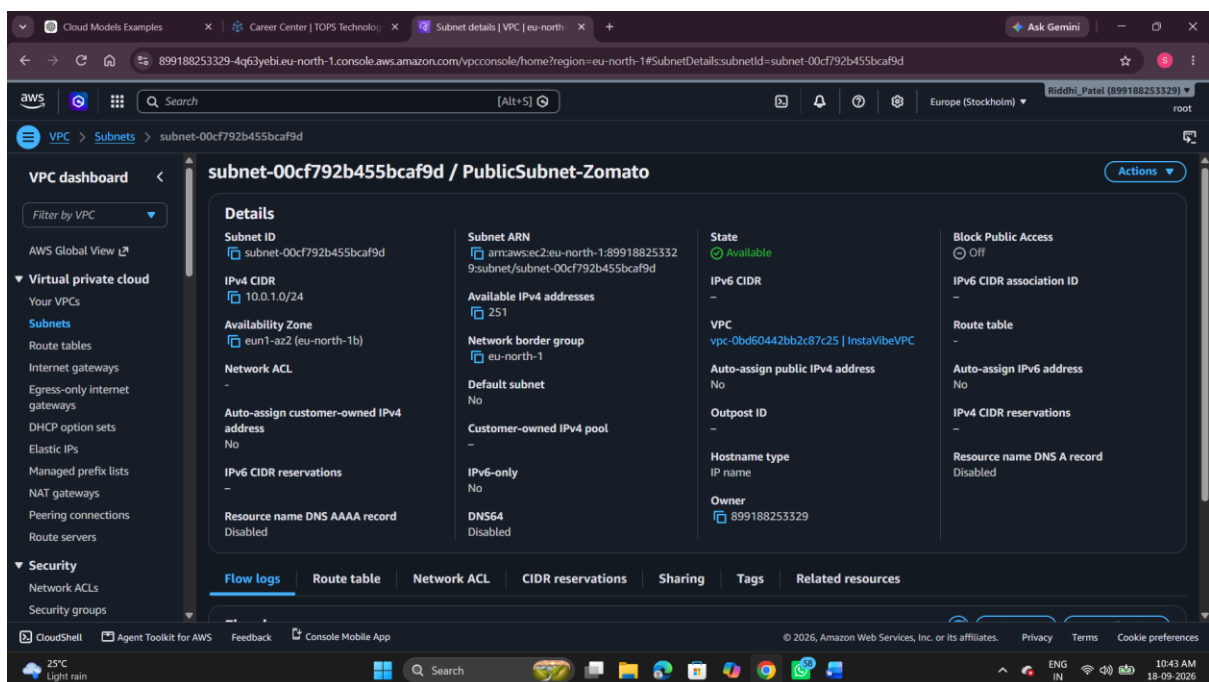
2. Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.1.0/24) for hosting a web server and one private subnet (e.g.,

10.0.2.0/24) for backend services. Hint: Name your subnets as 'PublicSubnet-Zomato' and 'PrivateSubnet-Zomato' for clarity.

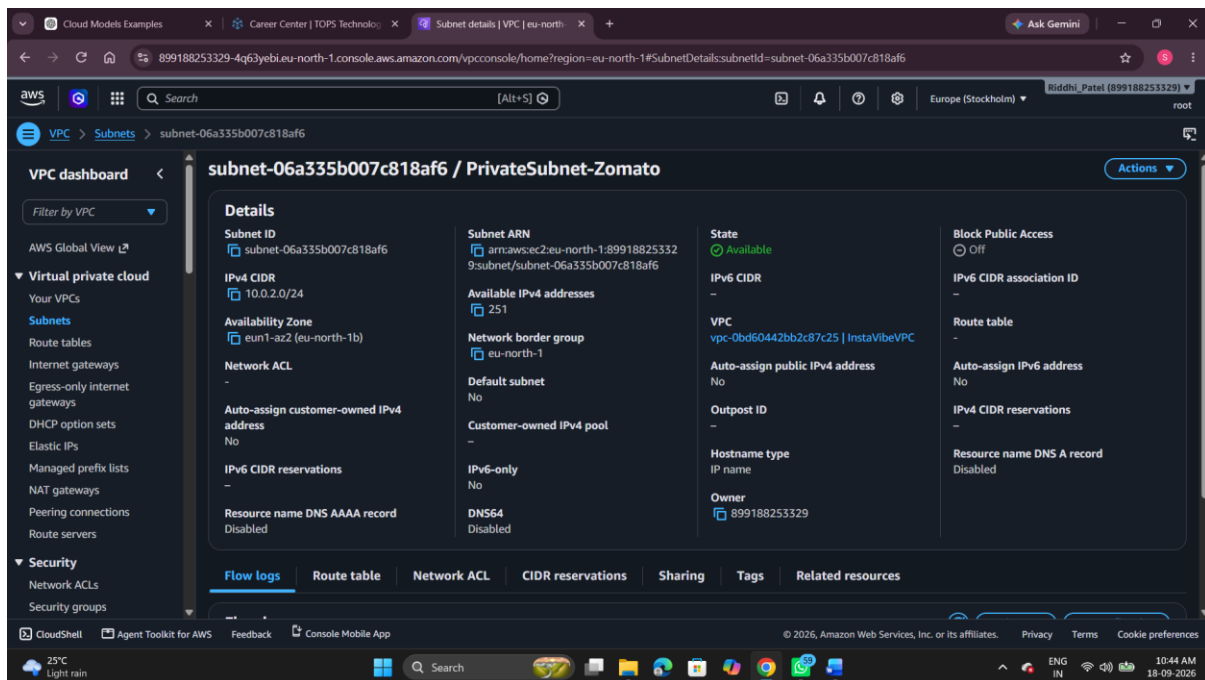
Ans : Steps:

1. Open AWS VPC Dashboard → Subnets.
2. Click Create subnet.
3. Select VPC: InstaVibeVPC.
4. Create the first subnet:
 - Name: PublicSubnet-Zomato
 - IPv4 CIDR: 10.0.1.0/24
5. Create the second subnet:
 - Name: PrivateSubnet-Zomato
 - IPv4 CIDR: 10.0.2.0/24
6. Click Create subnet.

Screenshot 1: PublicSubnet-Zomato showing CIDR 10.0.1.0/24



Screenshot 2: PrivateSubnet-Zomato showing CIDR 10.0.2.0/24



3. Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

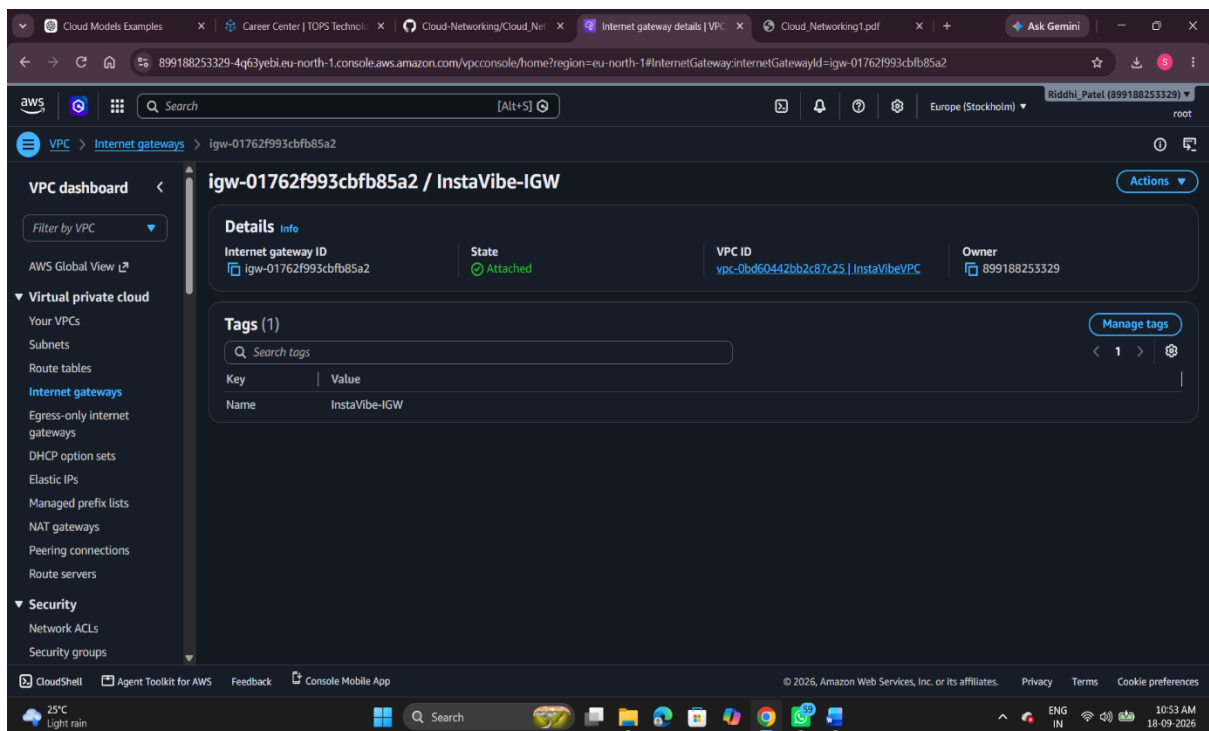
Ans:

I created an Internet Gateway and attached it to the InstaVibeVPC. I configured the route table so that the PublicSubnet-Zomato can access the internet, while the PrivateSubnet-Zomato does not have a direct internet route.

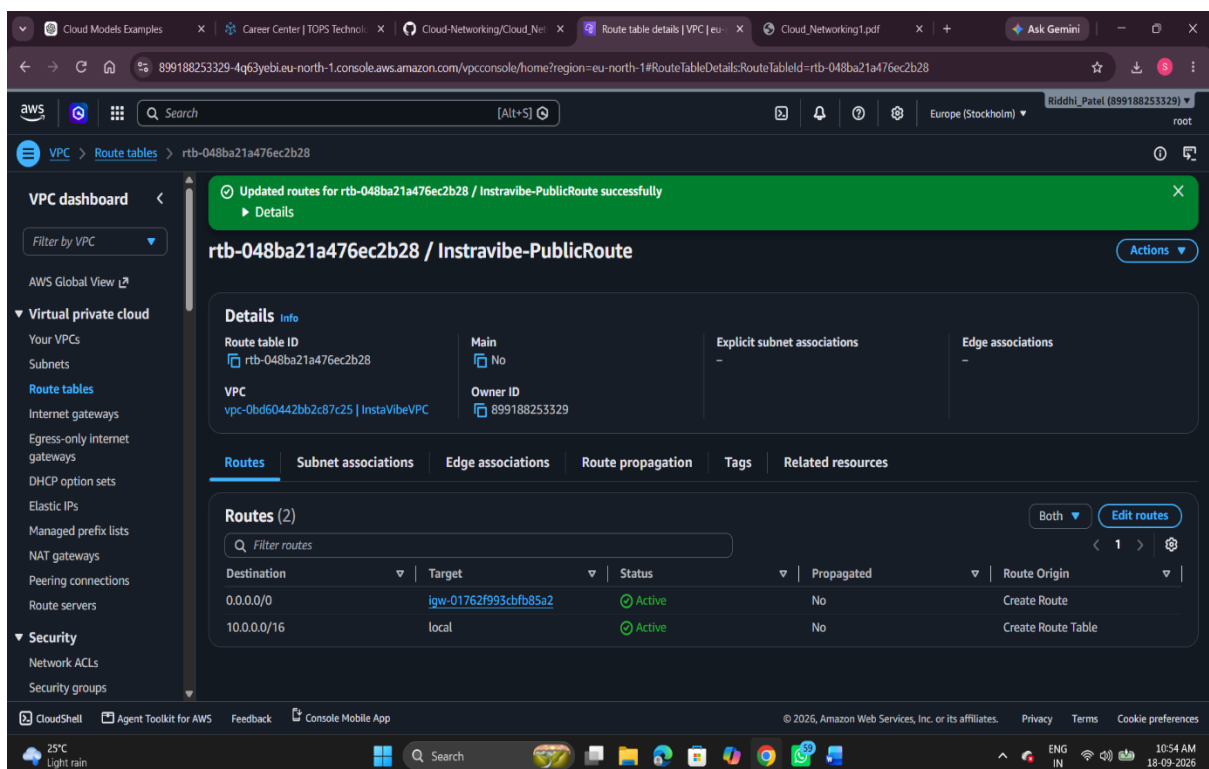
Configuration:

- Internet Gateway: InstaVibe-IGW
- VPC: InstaVibeVPC
- Internet Route: 0.0.0.0/0 → Internet Gateway
- Subnet associated: PublicSubnet-Zomato

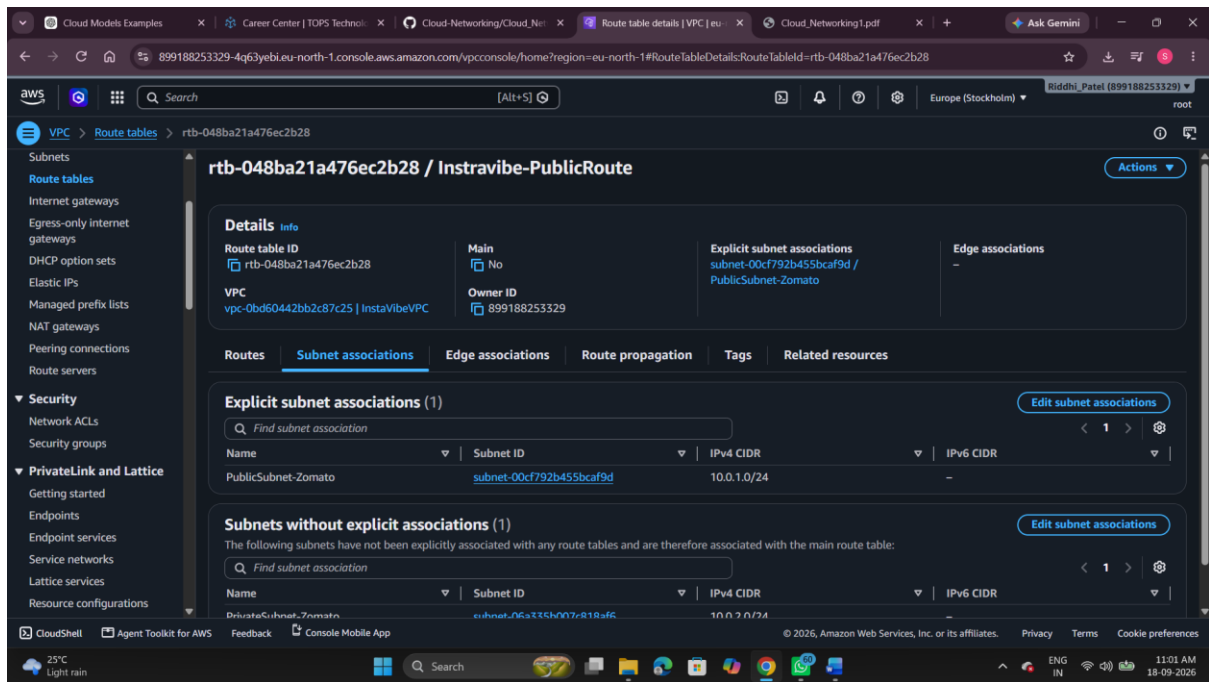
Screenshot 1: Internet Gateway attached to InstaVibeVPC.



Screenshot 2: Route table showing 0.0.0.0/0 → Internet Gateway.



Screenshot 3: Route table subnet association showing PublicSubnet-Zomato



Result:

The public subnet is configured for internet access, while the private subnet has no direct route to the Internet Gateway.

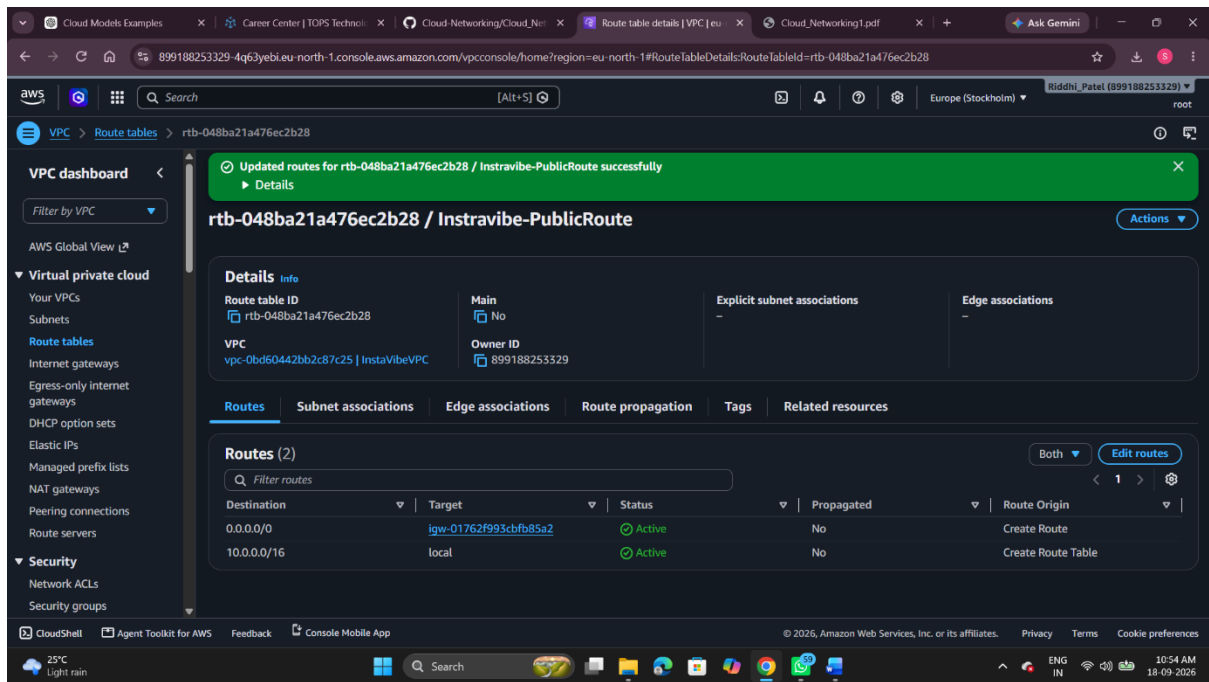
4. Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

Ans : I configured separate route tables for the public and private subnets.

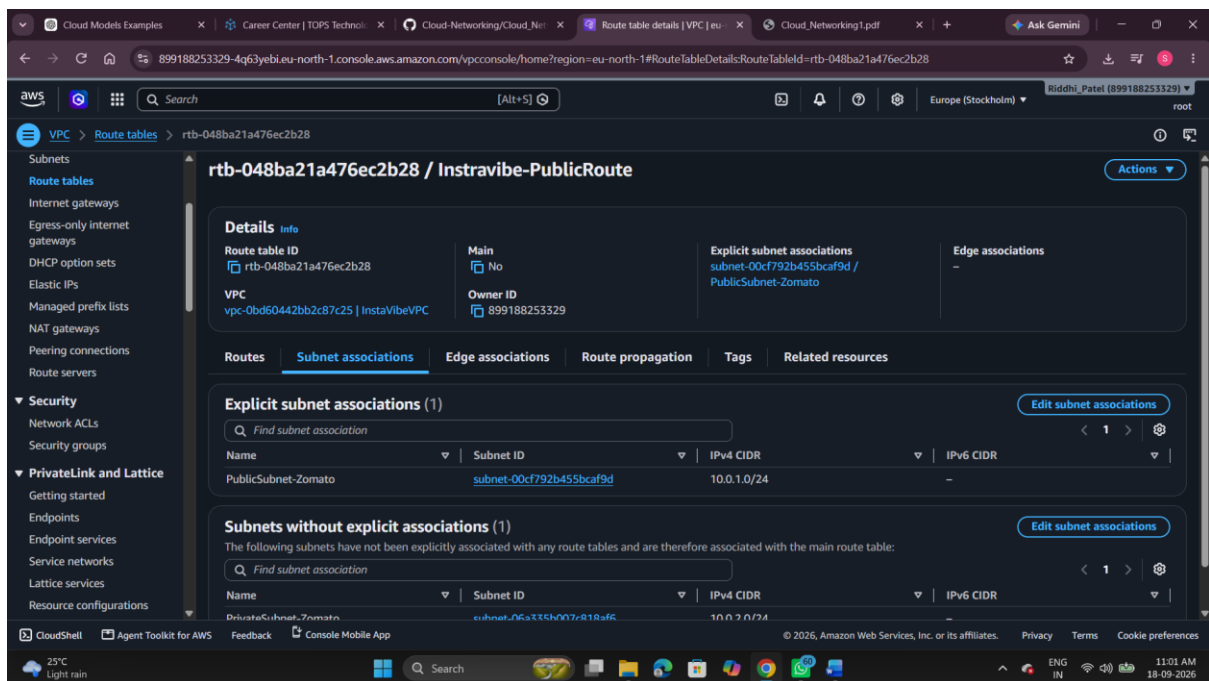
- The Public Route Table has a route 0.0.0.0/0 to the Internet Gateway and is associated with PublicSubnet-Zomato.
- The Private Route Table is associated with PrivateSubnet-Zomato and does not have a direct route to the Internet Gateway.

This ensures that the public subnet can access the internet, while the private subnet does not have direct internet access.

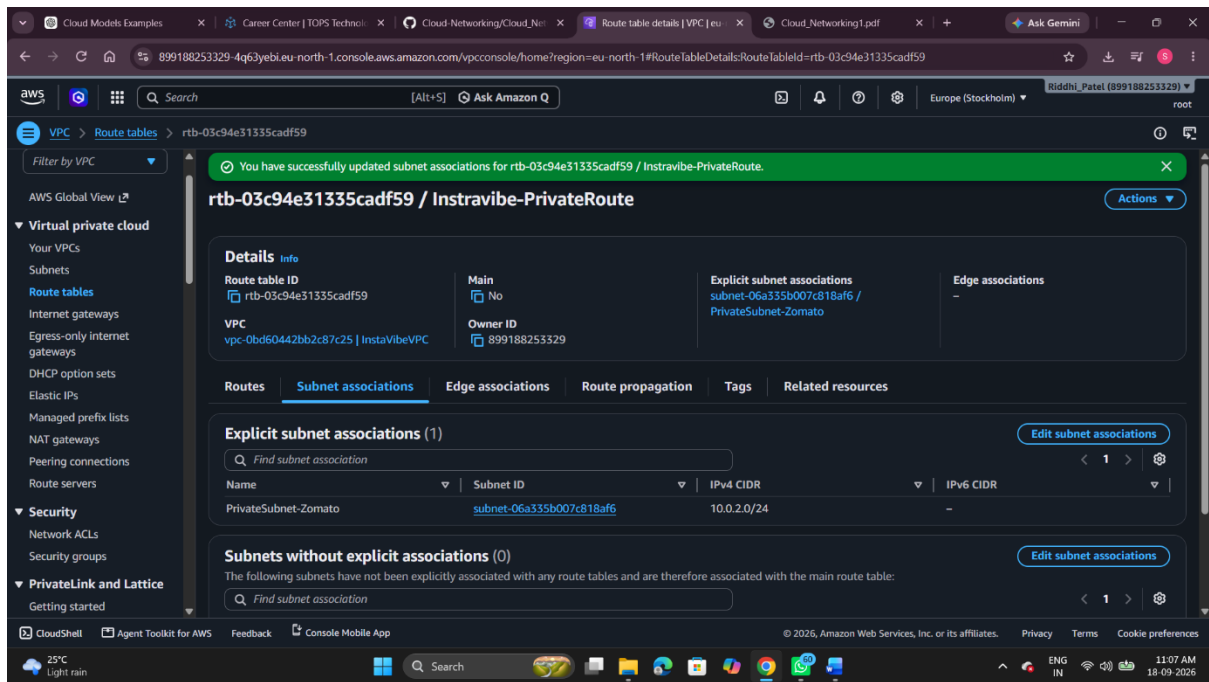
Screenshot 1: Public Route Table showing 0.0.0.0/0 → Internet Gateway



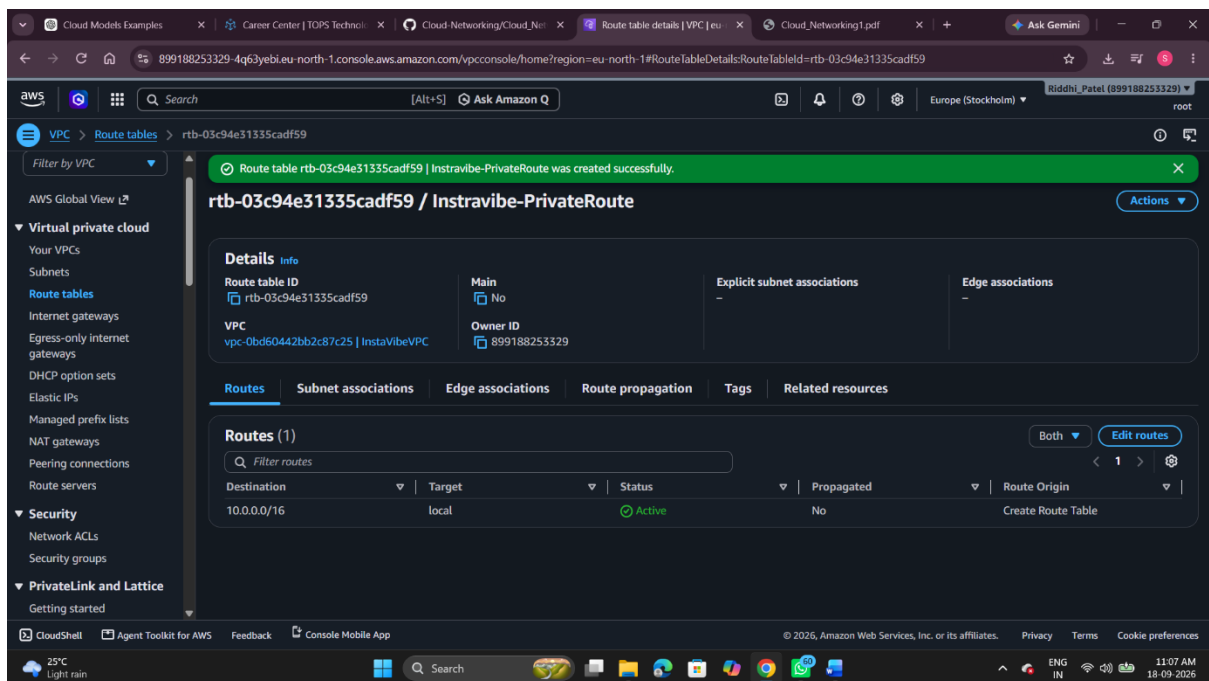
Screenshot 2: Public Route Table → Subnet Associations → PublicSubnet-Zomato



Screenshot 3: Private Route Table → Subnet Associations → PrivateSubnet-Zomato

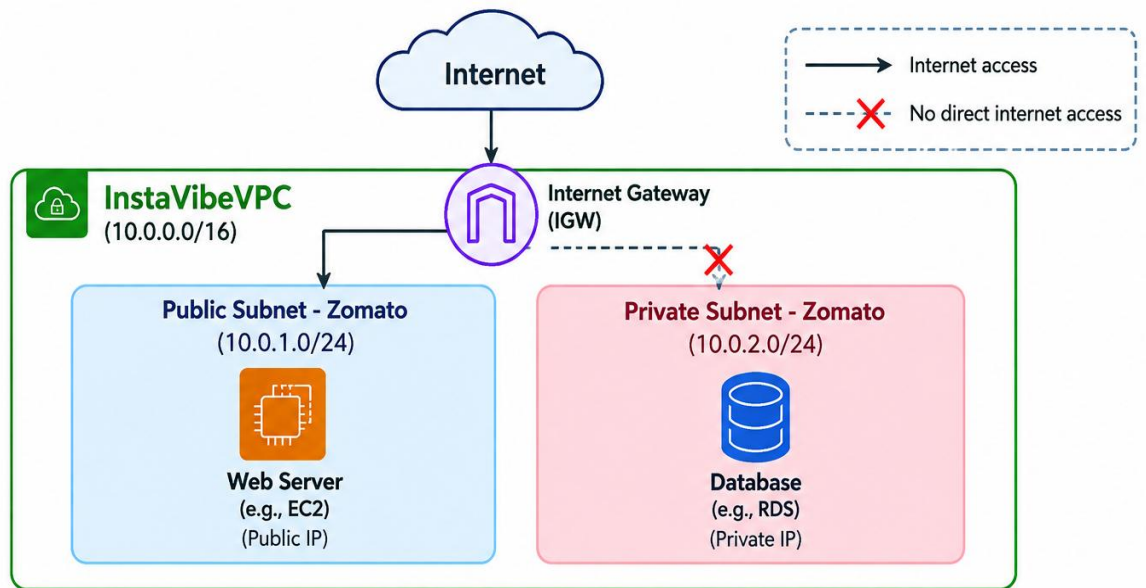


Screenshot 4: Private Route Table → Routes showing no Internet Gateway route.



5. Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.

Ans :



How this is similar to Zomato

1. Zomato's user interface (website/app) is like the public subnet, which is accessible from the internet so that users can browse restaurants, place orders, etc.
2. Zomato's databases and backend services are like the private subnet, which are not directly accessible from the internet for security.
3. This separation keeps sensitive data (like customer and order details) safe while still allowing users to access the application.